Conditional Reconstruction of a Quasi-Local Observable Domain within Compact S^3 Geometry

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We present a conditional reconstruction of a quasi-local observable domain inside a compact, simply connected, positively curved round three-sphere S^3_R. The reconstruction is not a topology detection, dark-sector derivation, or residual-source completion. It is a background-reconstructive and benchmark-ready compact-geometry framework. In standard FLRW convention, closed positive spatial curvature corresponds to $\Omega_K < 0$. We use the fiducial modulus $|\Omega_K| = 0.0007$ only as a curvature-radius benchmark and set $\Omega_K^{S^3} = -0.0007$ for the closed branch. With $H_0 = 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}}$ this gives $R_{\mathcal{S}^3} \simeq 548.3\,\mathrm{Gly}$; machine-precision benchmark values are kept only in the numerical table.

The exact geometry then supplies $D_c = \pi R$, $C = 2\pi R$, $V_{S^3} = 2\pi^2 R^3$, $K = R^{-2}$, and ${}^{(3)}R = 6R^{-2}$. The observable particle-horizon scale used in the benchmark, $D_{\mathrm{particle}} \simeq 46.13\,\mathrm{Gly}$, is treated as background dependent rather than as a theorem of S^3 geometry. Its geodesic-ball fraction is $\Gamma_{\mathrm{obs}} = 1.2614 \times 10^{-4} \simeq 0.0126\%$. The static-carrier comparison $ct_0/R \simeq 1.44°$ is only a diagnostic scale; FLRW causal propagation is governed by conformal distance. The finite radius is therefore a curvature scale, not a boundary: $\partial S^3_R = \varnothing$, while observer-defined subdomains can have relational reconstruction boundaries.

The predictive content is registered through the chain $|\Omega_K| \to R_c \to \{D_M, H, dV/dz\,d\Omega\} \to \{\beta, \Delta^X_{\ell\beta}\} \to \{C_\ell^{TT,TE,EE}, P(k,z), \xi\} \to \Delta \ln \mathcal{L}$ or $\Delta \ln Z$. Round S^3_R has no invariant spatial direction under the full $SO(4)$ symmetry; any axis, ring normal, or preferred frame must enter through data, fields, survey windows, initial conditions, or quotient topology S^3/Γ. Thus the paper's empirical target is not a preferred direction but the replacement of the flat continuum by the S^3 eigenvalue ladder, degeneracy law, addition kernel, windowed responses, and covariance matrices. All amplitude claims remain transfer- and likelihood-dependent.

I Introduction and Purpose

The aim of this paper is neither to prove that the physical Universe globally has the topology of S^3, nor to replace the standard cosmological model. The scope is deliberately narrower: we construct a conditional, but mathematically well-defined, reconstruction.

We take an observationally admissible small curvature modulus, together with the closed-sign interpretation $\Omega_K^{S^3} < 0$, not as a detected measurement but as a fiducial benchmark for a conditional positive-curvature reconstruction. In the standard FLRW convention,

$$\Omega_K < 0 \tag{1}$$

corresponds to a closed, positively curved spatial geometry. If the spatial section is modelled as a round three-sphere S^3_R, a large but finite curvature radius is recovered from the small modulus $|\Omega_K|$.

This approach continues the discipline already introduced in our earlier work [1]: S^3 is treated not as an arbitrary metaphor but as the minimal compact simply connected spherical carrier in which the following invariant structures meet: positive constant curvature, finite volume, a discrete round spectrum, $SU(2)$ spin geometry, and the Hopf fibration. Parallelisability alone is not sufficient to select S^3, since every closed orientable three-manifold is parallelisable [28]. The selection is therefore made by the full invariant bundle, not by a single property. The topology defines the carrier; dynamics, amplitudes, and physical fields require additional laws.

In the companion paper on light and causal coherence [1], the compact space $S^3_R \times \mathbb{R}$ is already used as the ambient model, where the causal diameter equals

$$D_c = \pi R, \tag{2}$$

and the global causal accessibility time is

$$t_{\mathrm{coh}} = \frac{\pi R}{c}. \tag{3}$$

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There it is emphasised that this is not an "echo of light", not superluminal communication, and not a mystical interpretation, but finite null accessibility within a compact, boundaryless space.

II Basic Reconstruction Hypothesis

Let the global spatial section be modelled as a round three-sphere:

$$\Sigma_3 = S^3_R. \tag{4}$$

Locally, for any sufficiently small observer domain:

$$U \subset S^3_R, \qquad U \simeq \mathbb{R}^3. \tag{5}$$

Therefore standard local physics is not abolished; it is preserved as a quasi-local approximation.

Globally, however,

$$S^3_R \neq \mathbb{R}^3. \tag{6}$$

The differences are essential:

- S^3_R is compact, without boundary;
- it has finite proper volume;
- it has a finite causal diameter;
- its Laplacian has a discrete spectrum;
- its maximal geodesic distance is πR.

In this sense the flat $\mathbb{R}^3$ framework is not the unique "default" ontology. It is itself a choice: non-compactness, infinite volume, and the absence of global closure. The compact S^3 framework offers a different choice: locally nearly flat physics within a globally finite geometry.

III Invariant Geometry of the Round Carrier

For the round three-sphere of radius R,

$$S^3_R = \{x \in \mathbb{R}^4 : \|x\| = R\}, \tag{7}$$

the intrinsic distance between two points is

$$d(x, y) = R \arccos\left(\frac{x \cdot y}{R^2}\right). \tag{8}$$

All local curvature invariants are fixed by the single radius R. The Riemann, Ricci, and scalar curvatures are

$$R_{ijkl} = \frac{1}{R^2}(g_{ik}g_{jl} - g_{il}g_{jk}), \tag{9}$$

$$R_{ij} = \frac{2}{R^2}g_{ij}, \tag{10}$$

$${}^{(3)}R = \frac{6}{R^2}. \tag{11}$$

The round carrier is locally symmetric:

$$\nabla_m R_{ijkl} = 0. \tag{12}$$

Consequently every local scalar curvature invariant is a polynomial in R^{-2}. For example,

$$R_{ijkl}R^{ijkl} = \frac{12}{R^4}, \tag{13}$$

$$R_{ij}R^{ij} = \frac{12}{R^4}, \tag{14}$$

$$\left({}^{(3)}R\right)^2 = \frac{36}{R^4}. \tag{15}$$

These identities complete the local invariant description of the round carrier: there is no hidden local curvature datum beyond the radius.

A No invariant spatial direction on round S^3

The round simply connected carrier has no nonzero invariant spatial direction. At any point $p \in S^3_R$, the stabilizer subgroup of the isometry group is $SO(3)$, acting on the tangent space T_pS^3. This action is transitive on the unit sphere in T_pS^3. If a tangent vector $v \in T_pS^3$ were invariant under all rotations in the stabilizer, then

$$h_* v = v \qquad \forall h \in SO(3), \tag{16}$$

which is possible only for

$$v = 0. \tag{17}$$

Thus a round S^3_R geometry by itself cannot generate a preferred axis, a preferred ring orientation, or a privileged spatial direction. In the same full $SO(4)$ sense, no nonzero invariant one-form or two-form is selected by the bare round carrier. Such features may arise only from data, observer position, survey masks, field dynamics, initial conditions, a Hopf-frame choice, or a nontrivial quotient topology S^3/Γ. The invariant content of the round carrier consists of distances, volumes, curvature, spectra, and correlation functions, not directions.

IV Invariant Selection Principle for S^3

The word "unique" is used in this paper only under a precise bundle of assumptions. Positive curvature by itself does not determine global topology: closed spherical space forms may be quotients S^3/Γ by finite freely acting isometry groups. Likewise, parallelisability by itself does not select S^3, since all closed orientable three-manifolds are parallelisable [28]. The strict selection statement is therefore conditional:

> **Invariant Selection Principle.** If the spatial carrier is required to be a smooth, compact, boundaryless, simply connected three-manifold with constant positive sectional curvature and with the minimal spinorial Lie-group carrier structure required by $SU(2)$, then the carrier is the round three-sphere S^3_R.

The invariant support chain is:

$$\text{closed} + \text{simply connected} + \text{dimension } 3 \Rightarrow S^3 \quad \text{(Poincaré–Perelman)}, \tag{18}$$

$$K > 0 \text{ and simply connected} \Rightarrow (S^3_R, g_R) \text{ as the spherical model}, \tag{19}$$

$$S^3 \simeq SU(2) \simeq Sp(1) \Rightarrow \text{natural spin/rotation double-cover carrier}, \tag{20}$$

$$S^1 \hookrightarrow S^3 \to S^2, \qquad [S^3 \to S^2] \in \pi_3(S^2) \simeq \mathbb{Z}. \tag{21}$$

where the last line is the nontrivial Hopf fibration. The theorem proved by Perelman, via Ricci flow and geometrization, states that every closed simply connected three-manifold is a three-sphere [25, 26]. The identification $S^3 \simeq SU(2)$ follows from unit quaternions and gives the standard double-cover relation $SU(2) \to SO(3)$ relevant for spin representations [29]. The Hopf fibration supplies a nontrivial S^1–fibered total space whose total space is S^3.

Thus S^3 is not claimed to be forced by observation alone. It is the unique object selected when the above invariant requirements are imposed simultaneously. This is the strict version of the topological claim used below.

V Comparison of Thurston Geometries

To illustrate the selection logic, Table I compares the eight Thurston model geometries appearing in geometrization. The table does not claim that no compact quotients exist in other geometries. Compact Euclidean and hyperbolic quotients, for example, can have discrete Laplacian spectra. The point is narrower: among the simply connected Thurston model carriers, only the spherical model simultaneously supplies compact closure, simple connectivity, constant positive sectional curvature, the $SU(2)$/Hopf structure, round spectral data, and native integer topological charges without imposing boundary conditions at infinity.

TABLE I. Thurston model geometries under the invariant bundle used in this reconstruction. Compact quotients of other model geometries may have discrete spectra, but they are not simply connected model carriers.

Geometry	Model	K	Compact SC?	$SU(2)$/Hopf?	Charges
Spherical	S^3	$+$	Yes	Yes	Yes
Euclidean	$\mathbb{E}^3$	0	No	No	quotient
Hyperbolic	$\mathbb{H}^3$	$-$	No	No	quotient
$S^2\times\mathbb{R}$	$S^2\times\mathbb{R}$	mixed	No	partial	no
$\mathbb{H}^2\times\mathbb{R}$	$\mathbb{H}^2\times\mathbb{R}$	mixed	No	No	no
$\widetilde{SL(2,\mathbb{R})}$	universal cover	mixed	No	partial	no
Nil	Heisenberg	mixed	No	No	no
Sol	solvable	mixed	No	No	no

A Relation to the Poincaré Dodecahedral Space

A useful comparison object is the Poincaré dodecahedral space,

$$M_{\mathrm{PDS}} = S^3/I^*, \tag{22}$$

where I^* is the binary icosahedral group of order 120. This is a compact multiply connected spherical space form of constant positive sectional curvature. Its fundamental domain may be represented by a regular dodecahedron whose opposite faces are identified after a 36° twist. Since the quotient has 120 copies in the simply connected cover, its volume at the same curvature radius is

$$V_{\mathrm{PDS}} = \frac{1}{120}V_{S^3} = \frac{2\pi^2R^3}{120}. \tag{23}$$

It is a homology three-sphere, but not a simply connected one:

$$H_*(M_{\mathrm{PDS}}) \simeq H_*(S^3), \qquad \pi_1(M_{\mathrm{PDS}}) \simeq I^* \neq 0. \tag{24}$$

The Poincaré dodecahedral space is historically important in CMB topology because it was proposed as a finite positively curved model capable of suppressing large-angle temperature correlations and reducing low multipole power [16, 18]. In such a quotient, the scalar harmonics are not the full set of round-S^3 harmonics but the subset invariant under the deck group I^*. Equivalently, let $\mathcal{H}^{(n)}_{S^3}$ denote the scalar eigenspace of $-\Delta_{S^3}$ with eigenvalue $n(n+2)/R^2$. For each such eigenspace one keeps only

$$\mathcal{H}^{(n)}_{\mathrm{PDS}} = \left\{ Y \in \mathcal{H}^{(n)}_{S^3} : \rho(g)Y = Y \ \ \forall g \in I^* \right\}, \tag{25}$$

so the multiplicity is the dimension of the trivial I^* representation inside the restricted S^3 eigenspace. In character language this is the group projection

$$m_n^{\mathrm{PDS}} = \frac{1}{|I^*|}\sum_{g\in I^*}\chi_n(g), \tag{26}$$

where χ_n is the character of the corresponding round-S^3 harmonic representation restricted to I^*. Thus the PDS spectrum is a group-projected subcatalogue of the S^3 spectrum, not a new local curvature spectrum. Published PDS CMB calculations often use the notation $\lambda_\beta = (\beta^2-1)/R^2$, related to the present convention by

$$\beta = n+1, \qquad \lambda_n = \frac{n(n+2)}{R^2} = \frac{\beta^2-1}{R^2}. \tag{27}$$

In this notation the first nonconstant surviving PDS scalar modes occur only at comparatively high β values, such as $\beta = 13$ [18]. This illustrates the stronger infrared gap and reduced low-mode degeneracy relative to the simply connected cover. The radius R and local curvature $K = 1/R^2$ are the same; the difference is the quotient projection on the allowed global modes.

The observational signature is also different. A multiply connected quotient may produce repeated images of the last-scattering surface and hence pairs of matched circles in the CMB sky [7, 17]. For the Poincaré dodecahedral model the standard signature consists of six pairs of matched circles with the characteristic dodecahedral phase shift. By contrast, the simply connected sector adopted here,

$$\Gamma = \{1\}, \qquad M = S^3_R, \tag{28}$$

has no nontrivial deck transformations and therefore no matched-circle signature of this quotient type. The absence of detected matched circles in Planck rules out some small fundamental domains but does not close the topology programme; current COMPACT analyses emphasize that topology searches remain computationally and statistically open [8, 9]. Recent lens-space circle searches also show how quotient sectors $L(p,q) = S^3/\mathbb{Z}_p$ are constrained by the absence of matched circles, while the simply connected sector has no such quotient-circle signature [11].

This distinction clarifies the role of the Invariant Selection Principle. The present paper does not claim that all spherical space forms are impossible. Rather, it deliberately selects the minimal simply connected carrier. Curvature alone fixes only the local spherical model and radius; it does not choose between S^3 and quotients S^3/Γ. The additional simply connected requirement sets $\Gamma = \{1\}$ and therefore selects the universal cover S^3_R.

Thus the Poincaré dodecahedral space is best understood here as the leading excluded quotient sector. If future CMB temperature and polarization searches were to detect the specific matched-circle or off-diagonal correlation pattern of a nontrivial quotient, the simply connected sector choice would have to be revised. If such signatures remain absent while low-ℓ anomalies persist, the large-radius simply connected compact-mode programme remains the cleaner minimal option.

For a general quotient $M = S^3/\Gamma$, the scalar eigenspace is

$$\mathcal{H}_\beta(M) = \{Q \in \mathcal{H}_\beta(S^3) : Q(gx) = Q(x)\ \forall g \in \Gamma\}, \tag{29}$$

with multiplicity

$$m^\Gamma_\beta = \frac{1}{|\Gamma|}\sum_{g\in\Gamma}\chi_\beta(g). \tag{30}$$

Such quotient sectors can generate off-diagonal CMB correlations of the form

$$C^\Gamma_{\ell m,\ell' m'} = \sum_{\beta,s}\mathcal{P}_\beta \Delta_{\ell\beta}\Delta_{\ell'\beta}\xi_{\beta s;\ell m}\xi^*_{\beta s;\ell' m'}. \tag{31}$$

For the simply connected round sector $\Gamma = \{1\}$, statistical isotropy gives

$$C_{\ell m,\ell' m'} = C_\ell \delta_{\ell\ell'}\delta_{mm'}. \tag{32}$$

Therefore any attempt to explain a persistent axis or orientation by topology alone must move from the round simply connected carrier to a quotient sector or to additional anisotropic dynamics.

TABLE II. Comparison between the simply connected round carrier used in this paper and the Poincaré dodecahedral quotient sector. At fixed curvature radius R, the quotient changes the allowed global modes and volume, not the local sectional curvature $K = 1/R^2$.

Feature	Simply connected carrier S^3_R	Poincaré dodecahedral space S^3/I^*
Topology	$\pi_1(S^3) = 0$	$\pi_1(S^3/I^*) \simeq I^*$, $\lvert I^*\rvert = 120$
Sector choice	Minimal universal cover, $\Gamma = \{1\}$	Nontrivial spherical quotient, $\Gamma = I^*$
Volume at same R	$V_{S^3} = 2\pi^2 R^3$	$V_{S^3}/120$
Spectrum	Full round-S^3 hyperspherical spectrum	I^*-invariant subset of S^3 modes
Low-mode effect	Subtle compact-mode transfer effect for large R	Stronger infrared gap from group projection at fixed R
CMB signature	No quotient matched-circle signature	Six matched-circle pairs with dodecahedral phase relation
Status in this paper	Selected by the simply connected invariant bundle	Excluded quotient sector / falsification alternative

VI Quantum-Scale Consequences of the Round Carrier

Global compactness becomes directly measurable only on scales comparable to the curvature radius $R_{S^3} \approx 548.32\,\mathrm{Gly}$. On all smaller laboratory, atomic, nuclear, and ordinary cosmological scales inside the particle horizon, every sufficiently small patch $U \subset S^3_R$ remains locally Euclidean, $U \simeq \mathbb{R}^3$, so the local form of standard physics is preserved. The quantum relevance of the round carrier is therefore not a local "new force". It appears through global spectral, spinorial, and topological data that are intrinsic to round S^3 rather than to an arbitrary three-manifold.

A Conservation: Strict Form and Limitation

For a compact spatial section without boundary there is no surface at spatial infinity through which a flux term can escape. In a static product spacetime $S^3_R \times \mathbb{R}$ with a timelike Killing field $X = \partial_t$, the Noether charge

$$E_X[\Sigma] = \int_\Sigma T_{\mu\nu} X^\mu n^\nu \, d\Sigma \tag{33}$$

is conserved under the usual stress-energy conservation law $\nabla_\mu T^{\mu\nu} = 0$. This is the correct strict statement. In a genuinely dynamical FLRW spacetime there is generally no global timelike Killing vector, so one should not claim an absolute globally conserved energy in the same sense; the invariant statement is local covariant conservation plus exact global charges only for actual symmetries.

Thus the role of S^3 is precise: compactness removes boundary-at-infinity losses, while exact global conservation requires the corresponding Killing symmetry. This correction prevents the argument from confusing compact topology with time-translation invariance.

B Round Spectral Data

On the round three-sphere the scalar Laplace–Beltrami spectrum is discrete. It is convenient to use the standard integer $\beta = n + 1$:

$$-\Delta_{S^3} Q_{\beta\ell m} = \frac{\beta^2 - 1}{R^2} Q_{\beta\ell m}, \qquad \beta = 1, 2, 3, \ldots. \tag{34}$$

For each β,

$$\ell = 0, 1, \ldots, \beta - 1, \qquad m = -\ell, \ldots, \ell, \tag{35}$$

and the scalar degeneracy is

$$g_\beta = \beta^2. \tag{36}$$

This is the same spectrum as

$$-\Delta Y_{n\ell m} = \frac{n(n+2)}{R^2} Y_{n\ell m}, \qquad g_n = (n+1)^2, \tag{37}$$

with $\beta = n + 1$. To avoid mixing conventions, the correspondence is

Old index	Hyperspherical index	Scalar eigenvalue	Degeneracy
$n = 0$	$\beta = 1$	0	1
$n = 1$	$\beta = 2$	$3/R^2$	4
$n = 2$	$\beta = 3$	$8/R^2$	9
n	$\beta = n + 1$	$n(n+2)/R^2 = (\beta^2 - 1)/R^2$	$(n+1)^2 = \beta^2$

The addition theorem gives the spectral kernel in closed form:

$$\sum_{\ell m} Q_{\beta\ell m}(x) Q^*_{\beta\ell m}(y) = \frac{\beta}{2\pi^2 R^3} \frac{\sin(\beta\gamma)}{\sin\gamma}, \tag{38}$$

$$\gamma = \frac{d(x,y)}{R}. \tag{39}$$

This is the invariant bridge between compactness and two-point correlations. A derivation of this addition theorem, its normalization, and its flat-space limit is given in Appendix A. The heat trace is correspondingly a sum, not a continuum integral:

$$Z(t) = \sum_{\beta=1}^{\infty} \beta^2 \exp\left[-t \frac{\beta^2 - 1}{R^2}\right]. \tag{40}$$

The difference from $\mathbb{R}^3$ is therefore not a local force; it is the replacement of a continuous k-integral by a discrete hyperspherical catalogue.

The spectrum of free fields on $S^3_R \times \mathbb{R}$ is a discrete mode problem rather than an infinite-volume continuum problem. This gives exact hyperspherical-harmonic expansions for scalar, vector, tensor, and spinorial fields and is the mathematical basis of a compact-mode CMB and vacuum-energy programme. Vacuum-energy sums are still infinite discrete sums and require a renormalization prescription such as zeta-function, heat-kernel, or physical subtraction. The improvement relative to $\mathbb{R}^3$ is the absence of an infinite-volume infrared continuum and the presence of exact degeneracies and mode gaps. Casimir calculations in Einstein-universe backgrounds illustrate precisely this role of curvature and topology in renormalized vacuum energy [43].

C Spinors and the Dirac Gap

For the round spin three-sphere, the Dirac spectrum is explicitly known; with a common sign convention one may write

$$\lambda_k^{\pm} = \pm\frac{k+3/2}{R}, \qquad k = 0, 1, 2, \ldots. \tag{41}$$

Thus the round geometry supplies a nonzero spectral gap of order R^{-1} for the untwisted Dirac operator [36]. The Lichnerowicz formula implies that a compact spin manifold with positive scalar curvature has no untwisted harmonic spinors [37]. This is a strong stability statement for the simplest spin sector.

The limitation is equally important: twisted Dirac operators, gauge backgrounds, or additional bundles can alter zero-mode questions. Therefore the article uses the Dirac gap as a controlled round-S^3 spectral fact, not as a universal proof that all fermionic sectors are automatically anomaly-free.

D Hopf Fibration and Quantum Phase Geometry

A normalized one-qubit state is a point on $S^3 \subset \mathbb{C}^2$ before quotienting by the physically irrelevant global phase. The Hopf fibration

$$S^1 \hookrightarrow S^3 \twoheadrightarrow S^2 \tag{42}$$

then sends the total state sphere to the Bloch sphere, while the circular fiber records the overall phase degree of freedom. This is the standard geometric relation between qubit state geometry, projective Hilbert space, and phase holonomy [32].

For field configurations the same domain S^3 supports the Hopf invariant:

$$[S^3, S^2] \simeq \pi_3(S^2) \simeq \mathbb{Z}. \tag{43}$$

The linking of inverse images under maps $S^3 \to S^2$ is the geometric content of this invariant, and it underlies Hopfions and knotted field configurations in Faddeev–Skyrme type models [33, 34]. On ordinary $\mathbb{R}^3$ this structure is normally obtained only after imposing a boundary condition at infinity and compactifying $\mathbb{R}^3 \cup \{\infty\} \simeq S^3$. In a genuine compact S^3 carrier the same topology is native rather than artificially added.

E Skyrmion Degree and Native Baryon Number

For the Skyrme field U, the target is $SU(2) \simeq S^3$. The topological charge is the degree of a map into $SU(2)$; in the usual $\mathbb{R}^3$ treatment this is well-defined after a boundary condition at infinity, effectively turning space into S^3 [35]. On a compact S^3 spatial carrier the same topological degree is intrinsic:

$$U : S^3 \to SU(2) \simeq S^3, \qquad B = \deg(U) \in \mathbb{Z}. \tag{44}$$

This supplies a precise sense in which a compact S^3 carrier makes Skyrmion number global without imposing an external point at infinity.

Claim type	Strict invariant content	Required wording in this article
Local physics	$U \subset S^3_R$, $U \simeq \mathbb{R}^3$	Standard local physics is preserved.
Global energy	No boundary-at-infinity flux; exact energy needs ∂_t Killing symmetry	Static conservation; local covariant conservation in FLRW.
Vacuum energy	Discrete spectrum and finite volume; UV sums still need renormalization	Computable renormalized mode sums, not finite bare sums.
Dirac sector	Positive scalar curvature; explicit round-S^3 gap	Untwisted spectral stability; twisted sectors separate.
Hopf/qubit sector	$S^1 \hookrightarrow S^3 \to S^2$, $\pi_3(S^2) = \mathbb{Z}$	Native phase, bundle, and Hopfion topology.
Skyrmion sector	$S^3 \to SU(2) \simeq S^3$, degree in $\mathbb{Z}$	Native global baryon-number topology.
Uniqueness	Full invariant bundle, not compactness alone	Conditional selection, not observational proof.

This section strengthens the article without overclaiming. The round S^3 carrier is not required in order to recover local quantum mechanics; it is selected by the demand that the global carrier simultaneously provide compact closure, exact round spectra, Hopf topology, $SU(2)$ spin geometry, and native integer topological charges.

VII Reconstructive Curvature Principle

The logical bridge from observation to geometry is formalised as follows.

Reconstructive Curvature Principle (RCP). *If the spatial curvature parameter satisfies $\Omega_K < 0$ within observational bounds, and if this value is treated not as noise but as a conditional signal of positive spatial curvature, then the standard FLRW curvature radius formula*

$$R_c = \frac{c/H_0}{\sqrt{|\Omega_K|}} \tag{45}$$

specifies the simply connected constant-positive-curvature universal cover $S^3_{R_c}$. The physical global section may in principle be a spherical space form S^3/Γ; the minimal simply connected sector sets $\Gamma = \{1\}$ and therefore selects $S^3_{R_c}$ itself.

Four points of discipline govern this principle:

1. **The RCP is conditional, not demonstrative.** It does not prove $\Omega_K < 0$ is real; it constructs what follows *if* it is.
2. **The sign is decisive.** The FLRW convention assigns $\Omega_K < 0$ to positive curvature (closed), $\Omega_K > 0$ to negative curvature (open), and $\Omega_K = 0$ to flat. A positive curvature reconstruction is therefore active only when $\Omega_K < 0$ is robustly non-zero.
3. **The formula is exact within the model.** Given H_0 and Ω_K, equation (45) is algebraically exact. Uncertainty enters only through the observational inputs.
4. **Topology requires a separate sector choice.** Curvature fixes the local spherical model and radius. It does not by itself exclude spherical quotient topologies S^3/Γ. The present paper uses the minimal simply connected carrier sector.

For the canonical Light-Trisphere reconstruction adopted in this paper, the input is not the Planck-only closed-universe anomaly. It is the fiducial radius-modulus benchmark

$$\begin{aligned} |\Omega_K| &= 0.0007, \\ \Omega_K^{S^3} &= -0.0007, \\ H_0 &= 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}}. \end{aligned} \tag{46}$$

The sign convention is essential: in standard FLRW notation $\Omega_K < 0$ means closed positive spatial curvature. Previous curvature-modulus estimates quoting $\Omega_K = +0.0007$ are therefore used here only through the same modulus $|\Omega_K| = 0.0007$, with the sign flipped for the closed S^3 interpretation. With these inputs,

$$R_c = \frac{c/H_0}{\sqrt{|\Omega_K|}} = 548.3\,\mathrm{Gly}, \tag{47}$$

yielding the fiducial radius adopted throughout this paper:

$$\boxed{R_{\mathcal{S}^3} = 548.3\,\mathrm{Gly}.} \tag{48}$$

A Degeneration at Zero Curvature

The limiting behaviour as $|\Omega_K| \to 0$ shows why a small non-zero value is categorically different from zero:

$$\begin{aligned} |\Omega_K| \to 0 &\Rightarrow R_c \to \infty \\ &\Rightarrow S^3_{R_c} \to \mathbb{R}^3 \quad \text{(locally and in the limit)}. \end{aligned} \tag{49}$$

The closed spherical model decompactifies into the flat limit. Conversely, even a tiny non-zero closed-sign curvature modulus produces a *finite* spherical curvature radius:

$$\begin{aligned} \Omega_K < 0 &\Rightarrow R_c < \infty \\ &\Rightarrow S^3_{R_c} \text{ is compact} \quad \text{(simply connected sector)}. \end{aligned} \tag{50}$$

For $\Omega_K > 0$ the corresponding FLRW sector is open/negative-curvature and belongs to the hyperbolic side, not to the S^3 reconstruction. The full reconstruction chain from observation to observable consequence is:

$$\begin{aligned} \Omega_K &\mapsto R_c \mapsto S^3_{R_c} \\ &\mapsto D_{\mathrm{obs}} \\ &\mapsto \text{modes, causal relations, horizons, structures.} \end{aligned} \tag{51}$$

A small closed-sign curvature is not "nearly zero" in the geometric sense; it fixes a finite radius for the spherical model. Zero gives decompactification; a closed-sign non-zero value gives a distinct compact spherical carrier in the chosen simply connected sector, with a specific spectrum, volume, and causal structure.

VIII Curvature-Distance Law and Reconstruction Test

The fiducial radius is only the kinematic starting point. Observable predictions require the FLRW distance-redshift relation, perturbation transfer functions, and a likelihood comparison to data. Let

$$E(z) = \frac{H(z)}{H_0}. \tag{52}$$

The line-of-sight comoving distance is

$$D_C(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')}. \tag{53}$$

For a closed spatial section with $\Omega_K < 0$, the dimensionless radial coordinate is

$$\chi(z) = \frac{D_C(z)}{R_c} = \sqrt{|\Omega_K|}\int_0^z \frac{dz'}{E(z')}. \tag{54}$$

The transverse comoving distance is therefore

$$\boxed{D_M(z) = R_c \sin\chi(z) = \frac{c/H_0}{\sqrt{|\Omega_K|}}\sin\left[\sqrt{|\Omega_K|}\int_0^z \frac{dz'}{E(z')}\right].} \tag{55}$$

The usual angular-diameter and luminosity distances are

$$D_A(z) = \frac{D_M(z)}{1+z}, \qquad D_L(z) = (1+z)D_M(z), \tag{56}$$

and the differential comoving volume per redshift and solid angle is

$$\frac{dV}{dz\,d\Omega} = \frac{c\,D_M^2(z)}{H(z)}. \tag{57}$$

For small curvature the expansion reads

$$D_M = D_C\left[1 + \frac{\Omega_K}{6}\left(\frac{H_0 D_C}{c}\right)^2 + O(\Omega_K^2)\right]. \tag{58}$$

Thus a closed model with $\Omega_K < 0$ gives a negative leading correction to the flat transverse distance at fixed D_C. At the particle-horizon scale used in this paper, $D_C \simeq 46.13\,\mathrm{Gly}$, the fiducial value $\Omega_K = -0.0007$ gives the order-of-magnitude diagnostic

$$\frac{D_M - D_C}{D_C} \simeq -1.179 \times 10^{-3}, \tag{59}$$

which is observable only through a full joint distance likelihood, not from the geometry alone.

A model-independent FLRW consistency reconstruction is

$$\boxed{\Omega_K(z) = \frac{[H(z)D_M'(z)]^2 - c^2}{H_0^2 D_M^2(z)}.} \tag{60}$$

For any FLRW geometry this quantity must be constant in redshift. The closed S^3 reconstruction requires, within observational errors,

$$\Omega_K(z) < 0, \qquad \frac{d\Omega_K}{dz} = 0. \tag{61}$$

This is a stronger observational test than selecting a single fiducial curvature modulus. In current data the curvature benchmark used here remains a permitted modulus rather than a detection: Planck 2018 combined with BAO gives a curvature constraint consistent with flatness [4], while DESI DR2 provides high-precision BAO distances from more than 14 million galaxies and quasars and primarily sharpens the distance-redshift likelihood rather than proving positive spatial curvature [13]. ACT DR6 extended-model constraints similarly report no statistically significant preference for departures from baseline ΛCDM, reinforcing the need to treat the present S^3_R construction as a registered likelihood programme rather than as a detected curvature sector [6].

A BAO and acoustic-scale diagnostic observables

The distance law enters present and future BAO analyses through the invariant combinations

$$D_H(z) = \frac{c}{H(z)}, \qquad D_V(z) = \left[z D_H(z) D_M^2(z)\right]^{1/3}, \tag{62}$$

$$F_{\rm AP}(z) = \frac{D_M(z) H(z)}{c}, \qquad \ell_A = \pi \frac{D_M(z_*)}{r_s(z_*)}. \tag{63}$$

At fixed line-of-sight distance D_C, the leading closed-curvature correction is

$$\frac{D_M - D_C}{D_C} = \frac{\Omega_K}{6}\left(\frac{H_0 D_C}{c}\right)^2 + O(\Omega_K^2), \qquad \Omega_K < 0. \tag{64}$$

Thus the purely geometric transverse correction is negative for the closed sector. For volume and acoustic observables, however, the measured shift also depends on which background expansion history $E(z)$ is compared. Therefore this paper does not register a standalone BAO number as a detection. The registered observable is the joint likelihood of

$$\{D_M/r_d,\ D_H/r_d,\ D_V/r_d,\ F_{\rm AP},\ \ell_A\} \tag{65}$$

under the same priors and covariance as the flat baseline.

B Compact mode-shift diagnostic

The compact spectrum samples wavenumbers at

$$k_\beta = \frac{\sqrt{\beta^2 - 1}}{R}, \tag{66}$$

rather than integrating continuously over k. For a smooth transfer integrand, the leading effect of replacing an integral by a sum may be diagnosed by the local spectral slope rather than asserted as a universal quadrupole suppression:

$$\Delta C_\ell^{XY} \sim \left.\frac{dC_\ell^{XY}}{d\ln k}\right|_{k=k_\ell} \Delta \ln k_\ell + \text{window/covariance terms}. \tag{67}$$

This expression is only a diagnostic expansion. The prediction used for data analysis remains the full hyperspherical transfer sum of Sec. XIX A and Sec. XXII.

IX Scope Guardrail: Background Expansion and Effective Acceleration

The reconstruction developed in this paper is geometric and spectral. It fixes a compact carrier, a curvature radius, the associated distance law, and a compact mode registry. It does not determine the physical origin of late-time acceleration. Consequently no statement in this article should be read as a proof that dark energy or dark matter do not exist, or as a derivation of a replacement for the late-time background sector.

Any effective acceleration component must enter only through a specified expansion history,

$$E(z) = \frac{H(z)}{H_0}, \tag{68}$$

and must be tested through the same BAO, supernova, CMB, lensing, and growth likelihoods used for standard cosmological models. The present work therefore uses $E(z)$ as an external background input for the distance and transfer-function pipeline. It does not import a separate dark-sector theory into the main reconstruction.

X Numerical Anchoring

In the main text we quote observationally meaningful rounded values. Machine-precision numbers are retained only as fiducial benchmark outputs in Table XXVI. For the adopted modulus one obtains

$$R_{\mathcal{S}^3} \simeq 548.3\,\mathrm{Gly}, \tag{69}$$

where $R_{\mathcal{S}^3}$ is the curvature radius of the global three-sphere.

In units of Gly/Gyr, the speed of light is numerically unity:

$$c = 1\,\mathrm{Gly/Gyr}. \tag{70}$$

Hence the light-travel time corresponding to the curvature radius is:

$$t_{\mathcal{S}^3} = \frac{R_{\mathcal{S}^3}}{c} \simeq 548.3\,\mathrm{Gyr}. \tag{71}$$

Important:

$$\boxed{t_{\mathcal{S}^3} \text{ is } \textit{not} \text{ the age of the Universe.}} \tag{72}$$

This is the light-travel time of the curvature radius. The age of the manifested hot cosmological history remains a separate quantity:

$$t_0 \simeq 13.79\,\mathrm{Gyr}. \tag{73}$$

XI Full Geometry of the Three-Sphere

A Diameter in the Embedding Space

If S^3_R is regarded as the standard sphere in $\mathbb{R}^4$,

$$S^3_R = \{x \in \mathbb{R}^4 : |x| = R\}, \tag{74}$$

the diameter in the ambient $\mathbb{R}^4$ embedding is:

$$\boxed{D_{\mathrm{embed}} = 2R = 1096.6\,\mathrm{Gly}.} \tag{75}$$

This is *not* the maximal intrinsic distance along the three-sphere itself.

B Intrinsic Causal Diameter $D_{\mathrm{causal}} = \pi R$

Two distinct diameter concepts must be carefully distinguished on S^3_R:

$$D_{\mathrm{embed}} = 2R \quad \text{(diameter in ambient } \mathbb{R}^4\text{)}, \tag{76}$$
$$D_{\mathrm{causal}} = \pi R \quad \text{(maximal intrinsic geodesic distance)}. \tag{77}$$

These differ by the factor $\pi/2 \approx 1.571$; conflating them leads to a systematic overestimate of the intrinsic scale by 57%.

On S^3_R, the maximum intrinsic geodesic distance between antipodal points is:

$$D_{\mathrm{causal}} \equiv D_{\mathrm{geo,max}} = \pi R. \tag{78}$$

Numerically:

$$\boxed{D_{\mathrm{causal}} = 1722.6\,\mathrm{Gly}.} \tag{79}$$

This is the causal diameter: the maximum finite distance any null signal can traverse within the compact spatial section. It is not a horizon in the particle-horizon sense; it is the intrinsic diameter of the manifold.

C Full Great Circle

The complete traversal along a great geodesic circle is

$$C = 2\pi R. \tag{80}$$

Numerically:

$$\boxed{C_{\mathcal{S}^3} = 3445.2\,\mathrm{Gly}.} \tag{81}$$

The corresponding light-travel time for a full traversal is

$$t_{\mathrm{loop}} = \frac{2\pi R}{c} = 3445.2\,\mathrm{Gyr}. \tag{82}$$

Half and quarter great circles are

$$\begin{aligned} \pi R &= 1722.6\,\mathrm{Gly}, \\ \frac{\pi R}{2} &= 861.3\,\mathrm{Gly}. \end{aligned} \tag{83}$$

XII Proper Volume of S^3

The proper three-dimensional volume of the round three-sphere is:

$$V_{S^3} = 2\pi^2 R^3. \tag{84}$$

Substituting $R = 548.3\,\mathrm{Gly}$ gives

$$\boxed{V_{\mathcal{S}^3} = 3.254 \times 10^9\,\mathrm{Gly}^3.} \tag{85}$$

This is *not* the volume of an ordinary ball, $V_{\mathrm{ball}} = 4\pi R^3/3$. It is the proper volume of the closed three-dimensional manifold S^3 itself.

XIII Curvature

For the round three-sphere, the sectional curvature is:

$$K_{\mathcal{S}^3} = \frac{1}{R^2}. \tag{86}$$

Numerically:

$$\boxed{K_{\mathcal{S}^3} = 3.326 \times 10^{-6}\,\mathrm{Gly}^{-2}.} \tag{87}$$

The three-dimensional scalar curvature is

$${}^{(3)}R = \frac{6}{R^2}. \tag{88}$$

Numerically:

$$\boxed{{}^{(3)}R = 1.996 \times 10^{-5}\,\mathrm{Gly}^{-2}.} \tag{89}$$

In SI units:

$$\boxed{|K|} = 3.716 \times 10^{-56}\,\mathrm{m}^{-2}, \tag{90}$$

$$\boxed{|{}^{(3)}R|} = 2.230 \times 10^{-55}\,\mathrm{m}^{-2}. \tag{91}$$

The curvature is positive but extremely small. A quasi-local observer will therefore consistently reconstruct their domain as nearly flat.

XIV Static-Carrier Age Diagnostic

The age of the hot cosmological history is

$$t_0 \simeq 13.79\,\mathrm{Gyr}. \tag{92}$$

Its ratio to the light-travel time of the curvature radius is

$$\frac{t_0}{t_{\mathcal{S}^3}} = \frac{13.788}{548.3}. \tag{93}$$

Evaluating:

$$\boxed{\frac{t_0}{t_{\mathcal{S}^3}} = 0.02515 \approx 2.515\%.} \tag{94}$$

The inverse ratio is

$$\boxed{\frac{t_{\mathcal{S}^3}}{t_0} = 39.77,} \tag{95}$$

i.e. the curvature-radius light-time exceeds the age of the observable hot history by a factor of approximately 39.77.

Expressing ct_0 as an arc on radius $R_{\mathcal{S}^3}$:

$$\theta_0 = \frac{ct_0}{R_{\mathcal{S}^3}} = \frac{t_0}{t_{\mathcal{S}^3}} = 0.02515\,\mathrm{rad}. \tag{96}$$

In degrees:

$$\boxed{\theta_0 = 1.441^\circ.} \tag{97}$$

Thus the static comparison length ct_0 corresponds to approximately 1.44° on a great circle of the fiducial carrier.

Note: This is a static-carrier diagnostic only. It is not the radius of the observable Universe and not the FLRW causal light-cone distance. In FLRW, causal propagation is controlled by the conformal distance $\int c\,d\tau/a(\tau)$, while the particle horizon is larger than ct_0 because of expansion.

XV Observable Horizon within the Three-Sphere

For the benchmark observable-domain calculation we adopt the particle-horizon radius

$$D_{\mathrm{particle}} = 46.13\,\mathrm{Gly}. \tag{98}$$

This value is not a theorem of S^3 geometry. It is a background-dependent input fixed by the chosen expansion history $E(z)$ and by the assumed radiation, matter, neutrino, and curvature content used to compute the conformal particle horizon. Its fraction of the three-sphere radius is

$$\frac{D_{\mathrm{particle}}}{R_{\mathcal{S}^3}} = 0.08412. \tag{99}$$

The corresponding angular size is

$$\theta_{\mathrm{particle}} = \frac{D_{\mathrm{particle}}}{R_{\mathcal{S}^3}} \cdot \frac{180^\circ}{\pi}. \tag{100}$$

Numerically:

$$\boxed{\theta_{\mathrm{particle}} = 4.820^\circ.} \tag{101}$$

Hence, the particle-horizon radius subtends approximately 4.82° on the great circle of S^3.

The diameter of the observable region is

$$2D_{\mathrm{particle}} = 92.25\,\mathrm{Gly}. \tag{102}$$

Its fraction of the full great circle is

$$\frac{2D_{\mathrm{particle}}}{2\pi R_{\mathcal{S}^3}} = 0.02678 \approx 2.678\%. \tag{103}$$

The central result of this section is that the observable Universe can be a small quasi-local domain of a large compact S^3.

A Volumetric Fraction of the Observable Domain

The angular subtended arc gives an intuitive picture, but the volumetric fraction is more quantitatively revealing. Define the geodesic half-angle:

$$\chi = \frac{D_{\text{particle}}}{R_{\mathcal{S}^3}} = 0.08412\,\text{rad}. \tag{104}$$

The fraction of the total S^3 volume enclosed within geodesic distance D_{particle} of the observer follows from the S^3 volume element:

$$\Gamma(\chi) = \frac{\chi - \frac{1}{2}\sin(2\chi)}{\pi}. \tag{105}$$

This follows directly from the round-S^3 volume element $dV = R^3 \sin^2\chi\, d\chi\, d\Omega_2$, since

$$V(\chi) = 4\pi R^3 \int_0^\chi \sin^2 u\, du = 2\pi R^3 \left(\chi - \tfrac{1}{2}\sin 2\chi\right), \tag{106}$$

and $V_{S^3} = 2\pi^2 R^3$. For small $\chi \ll 1$ (the regime relevant here), the leading-order expansion is:

$$\Gamma(\chi) \approx \frac{2}{3\pi}\chi^3 \qquad (\chi \ll 1). \tag{107}$$

Equation (107) shows that the volumetric fraction falls as the *cube* of the angular scale, making it far smaller than the linear angular comparison (4.82°) suggests. Substituting $\chi \simeq 0.08412$:

$$\begin{aligned} \Gamma_{\text{obs}} &= \Gamma(\chi_{\text{particle}}) \\ &= 1.261\,377\,743\,887\,7 \times 10^{-4} \\ &= 0.012\,613\,777\,438\,877\%. \end{aligned} \tag{108}$$

The small-angle approximation gives 0.012 631 642%, differing from the exact value by about 0.14% relative. Thus the observable Universe occupies about $\mathbf{1.26 \times 10^{-2}}$% of the total volume of S^3_R—not only angularly small, but volumetrically negligible. The corrected value follows directly from the cubic small-angle scaling in Eq. (107); the linear angular fraction must not be read as a volume fraction.

The $\Gamma(\chi)$ function is the spatial analogue of the causal fraction $\Gamma(T)$ introduced in Sec. XVIII; together they express the incompleteness of the observer's domain in both space and time. The joint benchmark reported in Table I is defined exactly as

$$\Gamma_{\text{ST}} = \Gamma\left(\frac{ct_0}{R}\right)\Gamma(\chi_{\text{particle}}) = 4.2551 \times 10^{-10}. \tag{109}$$

This is a static-carrier diagnostic product, not a new causal horizon in dynamical FLRW spacetime:

We observe a small quasi-local sector, not the global S^3.

XVI The Quasi-Local Observer Domain

A Formal Definition and Strict Incompleteness

The quasi-local observer domain is not identified with the global three-sphere. It is defined by finite causal accessibility, the particle horizon, the age of the manifested history, and the reconstruction procedure. Therefore, even under the compact S^3 ontology, the observable region remains a partial perspective.

Formally, the causal domain of an observer at point p after cosmic time T is:

$$D_{\text{obs}}(p,T) = \{q \in S^3_R : d_{S^3}(p,q) \le cT\}, \tag{110}$$

so that $D_{\text{obs}}(p,T) \subseteq S^3_R$. The strict incompleteness theorem holds as follows:

$$\begin{aligned} &D_{\text{obs}}(p,T) \subset S^3_R, \\ &D_{\text{obs}}(p,T) \neq S^3_R. \end{aligned} \tag{111}$$

This holds for every finite manifested reconstruction. This is not a contingent observational limitation. It is a structural consequence: a domain is defined by its observer and its horizon. Were it to coincide with the full

S^3_R, it would cease to be *quasi*-local and would instead be the global spatial section itself—a categorically different object.

The fractional causal accessibility volume is

$$\Gamma(T) = \frac{\mathrm{Vol}(D_{\mathrm{obs}}(p,T))}{\mathrm{Vol}(S^3_R)}. \tag{112}$$

For all $T < t_{\mathrm{coh}}$,

$$\Gamma(T) < 1. \tag{113}$$

B Boundary Without Boundary

The finite curvature radius recovered by the closed-branch reconstruction is not an edge of the Universe. It is a radius of curvature of the carrier geometry:

$$R_c = \frac{c/H_0}{\sqrt{|\Omega_K|}}, \qquad \Sigma \simeq S^3_{R_c} \tag{114}$$

for the simply connected closed sector. The global spatial carrier is compact and finite, but boundaryless:

$$\partial S^3_R = \varnothing. \tag{115}$$

By contrast, an observer-defined domain may have a nonempty relational boundary, for example

$$D_{\mathrm{obs}} \subset S^3_R, \qquad \partial D_{\mathrm{obs}} \simeq S^2. \tag{116}$$

Thus

$$\boxed{\partial S^3_R = \varnothing, \qquad \partial D_{\mathrm{obs}} \neq \varnothing, \qquad \partial D_{\mathrm{obs}} \not\equiv \partial S^3_R.} \tag{117}$$

The boundary of D_{obs} is a boundary of reconstruction, causal access, windowing, or imposed boundary data; it is not an ontological boundary of space.

This distinction is also the correct gravitational one. In covariant phase-space language the on-shell diffeomorphism current is exact, schematically

$$J_\xi \approx dQ_\xi, \tag{118}$$

so Hamiltonian generators for diffeomorphisms reduce to boundary functionals rather than bulk charges [41]. For a compact boundaryless global slice there is no spatial edge term. For a proper subregion $D \subset S^3_R$, however, the quasi-local generator is encoded on ∂D:

$$H_\xi[D] = \int_{\partial D} (Q_\xi - \xi \cdot B). \tag{119}$$

This is compatible with Brown–York type quasi-local energy constructions, but the present paper does not evaluate such charges or introduce a York-time dynamics sector [42]. The point used here is only categorical:

The universe has no global edge; observer domains may have quasi-local boundaries.

C Reachability Is Not Empirical Reconstruction

At $T = t_{\mathrm{coh}} = \pi R/c$, the condition $\Gamma(T) = 1$ is achieved—but only in the idealised static model sense of *null accessibility*:

$$\Gamma(t_{\mathrm{coh}}) = 1 \quad \Leftrightarrow \quad \begin{array}{l}\text{every point of } S^3_R \\ \text{is null-reachable from } p.\end{array} \tag{120}$$

This is a statement about geometric *reachability*—the existence of a null geodesic connecting any two points—not about empirical *reconstruction*. Reachability does not imply that an observer has access to, has measured, or has integrated the information from the entire S^3_R.

The distinction is essential:

$$\Gamma(t_{\mathrm{coh}}) = 1 \quad \Leftrightarrow \quad \text{global null accessibility (geometric fact)}, \tag{121}$$

$$D_{\mathrm{obs}}(p, t_{\mathrm{coh}}) = S^3_R \quad \not\Leftrightarrow \quad \text{full empirical reconstruction (physical claim)}. \tag{122}$$

The local null-cone structure is preserved at all times, and no single observer instantaneously integrates the entire compact section. The $\Gamma(T)$ function measures causal accessibility, not the actual data held by the observer.

D Protection against False Universalism

The two key methodological principles of this framework are:

> The quasi-local domain is the observational shadow of S^3, not S^3 itself.
> S^3 is not exhausted by any quasi-local perspective.

Standard flat cosmology performs the substitution locally reconstructed near-flatness $\to \mathbb{R}^3$-ontology. This programme refuses that substitution:

$$\begin{aligned} &\text{local near-flatness} \not\Rightarrow \mathbb{R}^3, \\ &\qquad\qquad \text{nor non-compact global geometry.} \end{aligned} \tag{123}$$

Local near-flatness is compatible with any large compact carrier: $U \subset S^3_R$, $U \simeq \mathbb{R}^3$, provided R is sufficiently large. The inference from local flatness to global non-compactness is an *additional ontological assumption*, not a deduction.

XVII Apparent Flatness and Curvature Sagitta Visualization

This section is an embedding visualization and curvature-intuition diagnostic only. It is not an invariant observational curvature estimator and does not replace the FLRW distance test in Eq. (60). To estimate the deviation from a flat chord, one may use the sagitta (arc-to-chord deviation):

$$s(L) = R - \sqrt{R^2 - \left(\frac{L}{2}\right)^2}, \tag{124}$$

which for small L/R reduces to

$$s(L) \approx \frac{L^2}{8R}. \tag{125}$$

For the light-travel age $L = ct_0 = 13.788\,\mathrm{Gly}$:

$$\boxed{s = 0.043\,338\,547\,\mathrm{Gly} = 43.34\,\mathrm{Mly}.} \tag{126}$$

For the particle horizon $L \simeq 46.13\,\mathrm{Gly}$:

$$\boxed{s = 0.485\,222\,026\,\mathrm{Gly}.} \tag{127}$$

For the diameter of the observable Universe $L = 92.250\,\mathrm{Gly}$:

$$\boxed{s = 1.943\,473\,550\,\mathrm{Gly}.} \tag{128}$$

These figures demonstrate: on the scale of the entire observable region, the curvature is geometrically non-zero, but it is extraordinarily gentle. Local reconstructions will therefore robustly yield a nearly flat picture.

XVIII Causal Accessibility

In the compact S^3_R geometry, the maximum intrinsic distance is $D_c = \pi R$, and the time of full causal accessibility is

$$t_{\mathrm{coh}} = \frac{\pi R}{c}. \tag{129}$$

For the chosen radius:

$$\boxed{t_{\mathrm{coh}} = 1722.6\,\mathrm{Gyr}.} \tag{130}$$

This is neither the "return time of a signal" nor a physical echo. It is the limiting time for null propagation to render every point of the spatial section causally accessible from every other point in a static compact

model. In the causal coherence framework [1], this is expressed through the compact causal diameter $D_c = \pi R$, the causal accessibility ratio $\Gamma(T)$, and the condition $\Gamma(T) = 1$ for $T \geq \pi R/c$.

The fraction of causally accessible volume at $0 < T < t_{\rm coh}$ is

$$\Gamma(T) = \frac{cT/R - \frac{1}{2}\sin(2cT/R)}{\pi}. \tag{131}$$

This function grows monotonically from 0 to 1. Thus, compact S^3 geometry has not infinite causal openness, but finite global causal accessibility.

XIX Research Programme

This calculation opens not a proof, but a research programme.

If the small negative Ω_K is accepted as a possible imprint of real positive spatial curvature, one can construct models not in infinite $\mathbb{R}^3$ but in compact S^3_R geometry with $R = 548.3\,\mathrm{Gly}$. The following directions are natural.

A CMB Low Multipoles and the Compact-Mode Test

The compact S^3_R reconstruction naturally motivates a dedicated analysis of the largest angular scales of the cosmic microwave background. The observed CMB temperature field exhibits several well-known low-multipole anomalies, including suppressed quadrupole power, quadrupole–octupole alignment, unusually weak two-point correlations at angular separations above roughly 60°, and related large-scale asymmetry or parity features [5, 23]. These anomalies are not regarded here as proof of nontrivial topology. Their statistical significance is modest and is affected by cosmic variance, foreground modelling, sky cuts, and look-elsewhere effects. They are therefore used only as phenomenological targets for a compact-mode test.

The correct predictive object is the closed-FLRW hyperspherical transfer spectrum, not the slogan that compactness suppresses the quadrupole. For $X, Y \in \{T, E\}$ the compact calculation should take the schematic form

$$\boxed{C_\ell^{XY,S^3} = \sum_\beta W_\beta\, \mathcal{P}_\mathcal{R}(\beta) \Delta^X_{\ell\beta} \Delta^Y_{\ell\beta}.} \tag{132}$$

The flat-space comparison is

$$C_\ell^{XY,\mathbb{R}^3} = 4\pi \int d\ln k\, \mathcal{P}_\mathcal{R}(k) \Delta^X_\ell(k) \Delta^Y_\ell(k). \tag{133}$$

The compact-mode prediction is the difference

$$\Delta C_\ell^{XY} = C_\ell^{XY,S^3} - C_\ell^{XY,\mathbb{R}^3}. \tag{134}$$

A likelihood-level test should use the data covariance:

$$\Delta\chi^2 = \sum_{\ell\ell',XY} \Delta C_\ell^{XY}\, \mathrm{Cov}^{-1}_{\ell XY,\ell' X'Y'} \Delta C_{\ell'}^{X'Y'}. \tag{135}$$

This is the operative CMB prediction pipeline.

For the fiducial radius used in this paper,

$$R_{\mathcal{S}^3} = 548.3\,\mathrm{Gly}, \tag{136}$$

the observed particle-horizon radius subtends only

$$\chi_{\rm obs} = \frac{D_{\rm particle}}{R_{\mathcal{S}^3}} = 0.08412\,\mathrm{rad} \simeq 4.82°. \tag{137}$$

Thus the observable region is a small quasi-local patch of the global carrier. The low multipoles cannot be claimed to be suppressed by topology alone without an explicit transfer calculation. Evidence in favour of the compact-mode interpretation would consist not merely of low quadrupole power, but of a coherent pattern across TT, TE, and EE, together with phase or correlation structures characteristic of the selected sector. Temperature-only anomalies are therefore insufficient; polarization is decisive [15].

B Large-Scale Structure

One may investigate whether a compact, gently curved geometry creates preferential conditions for large arcs, rings, walls, and correlation structures beyond the standard homogeneity scale. The formal chain is:

$$\Omega_K \mapsto R_{\mathcal{S}^3}^{\text{global}} \mapsto (S^3, g_R) \mapsto \Delta_{S^3} \mapsto Y_{n\ell\mu} \mapsto \text{projected shells} \mapsto \text{rings/arcs}. \tag{138}$$

C Compact-Mode vs. Box-Mode Simulations

A periodic Euclidean box T^3 is not equivalent to S^3. They have different modes, different spectral bases, and different global geometry. Simulations on S^3 harmonics can therefore produce different infrared effects.

D Matter and Gravity Dynamics

The next step is to specify the dynamics: FLRW evolution on S^3, perturbation theory, transfer functions, structure growth, lensing, and the CMB angular power spectrum. These are dynamical tasks, not consequences of topology alone. The compact carrier supplies the background geometry and spectral catalogue; it does not by itself determine amplitudes, matter content, or an effective acceleration sector.

E Causal Accessibility Observables

Through $\Gamma(T)$, $D_c = \pi R$, and compact null accessibility, one can search not for "echoes" but for correlation signatures of finite global causal accessibility.

F Global/Local Spectral Bridge to Ring-Like Ultra-Large-Scale Structures

In [2] we identified ring-like and arc-like structures at $z \approx 0.8$ with an effective spectral coherence scale $R_{\text{eff}} \approx 474.5\,\text{Mpc} \approx 1.547\,\text{Gly}$. The present curvature reconstruction requires a strict separation between two scales that must not be conflated:

$$\boxed{R_{\mathcal{S}^3}^{\text{global}} \neq R_{\text{eff}}.} \tag{139}$$

Global carrier: $R_{\mathcal{S}^3} \simeq 548.3\,\text{Gly}$ is the curvature radius of the three-sphere itself. It sets the Laplacian eigenvalues, the discrete mode spectrum, and the compact causal diameter. It is a property of the manifold.

Local spectral coherence scale: $R_{\text{eff}} \approx 474.5\,\text{Mpc}$ is an effective scale extracted from projected-shell correlations within a narrow redshift bin. Formally,

$$R_{\text{eff}} = R_{\text{eff}}(z, n, \Delta z, \mathcal{P}), \tag{140}$$

where $\mathcal{P}$ denotes the projection and selection kernel of the observed shell. It is a property of a particular S^3 mode projected onto the observable redshift shell; it depends on the shell's comoving distance, width, hyperspherical harmonic index, and survey window. This paper does not derive R_{eff}; it states the geometric separation between the global carrier radius and a projected local coherence scale.

The logical chain connecting the two levels is:

$$\begin{aligned} \Omega_K &\mapsto R_{\mathcal{S}^3}^{\text{global}} \mapsto (S^3, g_R) \mapsto \Delta_{S^3} \\ &\mapsto Y_{n\ell\mu} \mapsto \text{projected modes} \\ &\mapsto R_{\text{eff}}(z, n, \Delta z, \mathcal{P}). \end{aligned} \tag{141}$$

The 474.5 Mpc scale is therefore not the radius of the Universe but a *derived observable* of a specific compact mode in a specific shell. The global S^3 carrier provides the mode catalogue; the local R_{eff} is a read-out of one mode in one shell. As a first diagnostic, Sec. XXII records the projected-shell estimator $\beta_{\text{shell}} \simeq 618.38$ for $z = 0.8$ under the physical-scale convention stated in Sec. XXII. This is a target for the full shell-transfer pipeline, not a proof of a mode assignment by itself.

Before the present paper, the S^3 framework explained the *morphology* of ring-like structures. The curvature reconstruction now adds an *external geometric criterion*: the global carrier is conditionally recoverable from Ω_K, not just postulated for spectral convenience. In short:

> Before: S^3 explained ring morphology.
>
> Now: Ω_K provides the reconstructable background for that morphology.

XX Canonical Reconstruction Summary

A Three-Sphere Geometry

$$\begin{aligned} \mathcal{S}^3_{\text{Light}}: \quad R &= 548.3\,\text{Gly}, && (142)\\ t_R &= 548.3\,\text{Gyr}, && (143)\\ D_{\text{embed}} = 2R &= 1096.6\,\text{Gly}, && (144)\\ D_{\text{causal}} = \pi R &= 1722.6\,\text{Gly}, && (145)\\ C = 2\pi R &= 3445.2\,\text{Gly}, && (146)\\ V &\simeq 3.254\times 10^9\,\text{Gly}^3. && (147) \end{aligned}$$

B Manifested History

$$\begin{aligned} t_0 &= 13.788\,\text{Gyr}, && (148)\\ \frac{t_0}{t_R} &= 0.02515, && (149)\\ \theta_0 &= 1.441^\circ. && (150) \end{aligned}$$

C Observable Horizon

$$\begin{aligned} D_{\text{particle}} &= 46.13\,\text{Gly}, && (151)\\ \theta_{\text{particle}} &= 4.820^\circ, && (152)\\ \Gamma_{\text{obs}} &= \Gamma(\chi_{\text{particle}}) \approx 0.0126\%. && (153) \end{aligned}$$

D Fundamental Geometric Relations

The following relations hold as exact theorems of S^3_R geometry:

$$\begin{aligned} D_{\text{causal}} = \pi R, \quad V_{S^3} &= 2\pi^2 R^3,\\ K = \frac{1}{R^2}, \quad {}^{(3)}R &= \frac{6}{R^2}. \end{aligned} \tag{154}$$

Two-scale distinction (never conflate):

$$\begin{aligned} D_{\text{embed}} &= 2R \quad (\text{in } \mathbb{R}^4),\\ D_{\text{causal}} &= \pi R \quad (\text{intrinsic on } S^3). \end{aligned} \tag{155}$$

XXI Invariant Registry, Diagnostics, and Necessity Taxonomy

This section consolidates the invariant content needed for the reconstruction. It is not an encyclopedia of natural phenomena. It is a registry of the structures actually used by this paper.

A Exact round-S^3_R invariants

The round carrier is homogeneous and isotropic:

$$S^3_R \simeq SO(4)/SO(3), \qquad \text{Isom}^+(S^3_R) \simeq SO(4). \tag{156}$$

The stabilizer $SO(3)$ acts transitively on the unit sphere in each tangent space. Therefore there is no nonzero fully invariant direction:

$$\begin{aligned} \mathfrak{X}(S^3_R)^{SO(4)} &= 0, \qquad \Omega^1(S^3_R)^{SO(4)} = 0, \qquad \Omega^2(S^3_R)^{SO(4)} = 0,\\ \Omega^0(S^3_R)^{SO(4)} &= \mathbb{R}, \qquad \Omega^3(S^3_R)^{SO(4)} = \mathbb{R}\,\text{vol}_{S^3}. \end{aligned} \tag{157}$$

Equivalently, invariant directions, axes, ring normals, Hopf-frame directions, and preferred frames do not follow from the round simply connected carrier alone. They can only enter after additional structure is chosen: an observer, field configuration, subgroup, quotient topology, survey window, or initial condition. The same isotropy also fixes the invariant symmetric-tensor sector:

$$\text{Sym}^2(T^*S^3_R)^{SO(4)} = \mathbb{R}\,g. \tag{158}$$

Thus the round carrier alone cannot generate a geometric anisotropic stress tensor; any anisotropy must come from perturbations, matter fields, quotient topology, survey selection, or initial data.

The local curvature catalogue is completely fixed by R:

$$\begin{aligned} R_{ijkl} &= R^{-2}(g_{ik}g_{jl} - g_{il}g_{jk}), \\ R_{ij} &= 2R^{-2}g_{ij}, \qquad {}^{(3)}R = 6R^{-2}, \qquad \nabla_m R_{ijkl} = 0. \end{aligned} \tag{159}$$

The scalar curvature polynomials are therefore powers of R^{-2}, for example

$$R_{ijkl}R^{ijkl} = 12R^{-4}, \qquad R_{ij}R^{ij} = 12R^{-4}. \tag{160}$$

In three spatial dimensions the Weyl tensor has no independent components; the conformal-flatness diagnostic is the Cotton tensor. For the round carrier,

$$C^{\mathrm{Cotton}}_{ijk} = 0, \tag{161}$$

so the intrinsic spatial curvature is fully fixed by the metric and the single radius R. This statement is purely spatial and does not assert that the four-dimensional perturbed FLRW spacetime has vanishing Weyl curvature.

B Compact spectral diagnostics

The empirical content of S^3_R mode discreteness is the replacement of a flat continuum by a compact hyperspherical catalogue,

$$\int d^3k \quad \longrightarrow \quad \sum_{\beta\ell m}, \tag{162}$$

with

$$\begin{aligned} -\Delta Q_{\beta\ell m} &= \frac{\beta^2 - 1}{R^2} Q_{\beta\ell m}, \\ g^{(0)}_\beta &= \beta^2, \qquad k_\beta = \frac{\sqrt{\beta^2 - 1}}{R}. \end{aligned} \tag{163}$$

For the fiducial benchmark,

$$\begin{aligned} R^{-1} &= 1.8237 \times 10^{-3}\,\mathrm{Gly}^{-1}, \\ k_{\beta=2} &= 3.1588 \times 10^{-3}\,\mathrm{Gly}^{-1}. \end{aligned} \tag{164}$$

The constant mode $\beta = 1$ is a background mode and carries no mean-zero spatial structure. The first non-constant scalar mode is therefore $\beta = 2$, with

$$k^{(0)}_{\mathrm{min}} = \frac{\sqrt{3}}{R}. \tag{165}$$

This is an admissibility floor in the compact scalar catalogue. It is not, by itself, an amplitude-suppression theorem: any statement about quadrupole power or large-angle correlations still requires a primordial spectrum, transfer functions, survey window, and likelihood comparison. The exact scalar addition kernel is

$$K_\beta(\gamma) = \frac{\beta}{2\pi^2 R^3} \frac{\sin(\beta\gamma)}{\sin\gamma}, \qquad \gamma = d(x,y)/R. \tag{166}$$

The corresponding scalar correlation diagnostic is

$$\xi_{S^3}(\gamma) = \sum_{\beta=1}^{\infty} P_\beta \frac{\beta}{2\pi^2 R^3} \frac{\sin(\beta\gamma)}{\sin\gamma}. \tag{167}$$

The addition kernel obeys the normalization rule

$$\int_{S^3_R} K_\beta(\gamma(x,y))\, dV_y = \begin{cases} 1, & \beta = 1, \\ 0, & \beta > 1, \end{cases} \tag{168}$$

which separates the constant mode from all mean-zero modes. The high-mode counting is controlled by Weyl's law,

$$N(\Lambda) = \sum_{(\beta^2-1)/R^2 \le \Lambda} \beta^2 \sim \frac{R^3}{3}\Lambda^{3/2} \qquad (\Lambda \to \infty). \tag{169}$$

Consequently compact spectral effects are infrared diagnostics; at large β the compact sum approaches the flat continuum density. A useful angular scale for the particle-horizon benchmark is

$$\ell_{\mathrm{IR}}^{\mathrm{diag}} \sim \frac{1}{\chi_{\mathrm{particle}}} = \frac{R}{D_{\mathrm{particle}}} \simeq 11.9, \tag{170}$$

but this is only a diagnostic window, not a substitute for a full hyperspherical transfer calculation.

On a finite survey or causal domain D, the relevant quantity is the windowed response

$$W_\beta[D] = \frac{1}{\mathrm{Vol}(D)^2}\int_D dV_x \int_D dV_y\, K_\beta(\gamma(x,y)). \tag{171}$$

More generally, the finite observer domain induces a quasi-local mode-coupling matrix

$$\begin{aligned} M_{ab}[D] &= \int_D Q_a(x) Q_b^*(x)\, dV_x, \\ a &= (\beta,\ell,m), \qquad b = (\beta',\ell',m'). \end{aligned} \tag{172}$$

For $D = S^3_R$ this reduces to the orthonormal identity matrix. For a strict subdomain, including the particle-horizon ball or a real survey mask, $M_{ab}[D]$ need not be diagonal. This is a windowing and projection effect, not a new physical force. It can mix global compact modes in quasi-local estimators, but it does not by itself force a quadrupole deficit, an octupole alignment, or a preferred axis. Those claims require P_β, transfer functions, foreground and mask modelling, and a likelihood test. Thus compact-mode claims become observationally meaningful only after applying the survey mask, transfer functions, and likelihood.

The untwisted Dirac spectrum is also discrete,

$$\lambda_k^{\pm} = \pm\frac{k+3/2}{R}, \qquad d_k = (k+1)(k+2), \tag{173}$$

which gives the benchmark infrared spinor floor

$$\Delta_D = 5.7058 \times 10^{-35}\,\mathrm{eV}. \tag{174}$$

This is a mathematical infrared threshold of the round spin carrier, not a neutrino-mass prediction.

C Necessity taxonomy

The paper uses three necessity classes. First, structures native to the full round simply connected carrier include

$$S^3 \simeq SU(2), \qquad S^1 \hookrightarrow S^3 \to S^2, \qquad \pi_3(S^2) \simeq \mathbb{Z}, \tag{175}$$

and the Skyrme degree

$$U : S^3 \to SU(2) \simeq S^3, \qquad \deg(U) \in \mathbb{Z}. \tag{176}$$

Second, compactness-dependent but non-exclusive effects include discrete spectra, finite-volume quantum-field-theory corrections, heat kernels, and Casimir-type renormalized sums. These also occur on other compact manifolds or in bounded boxes. Third, ordinary local physics—local quantum mechanics, chemistry, local QFT, local GR, and laboratory-scale physics—does not require global S^3. It is preserved because $U \subset S^3_R$ is locally $\mathbb{R}^3$.

XXII Transfer and Likelihood Pipeline

The distance sector is the first observational layer. With $E(z) = H(z)/H_0$ supplied by an external background expansion model,

$$\chi(z) = \sqrt{|\Omega_K|} \int_0^z \frac{dz'}{E(z')}, \qquad D_M(z) = R_c \sin \chi(z). \tag{177}$$

The reconstruction test is

$$\Omega_K(z) = \frac{[H(z) D'_M(z)]^2 - c^2}{H_0^2 D_M^2(z)}. \tag{178}$$

A closed S^3_R reconstruction requires $\Omega_K(z) < 0$ and $d\Omega_K/dz = 0$ within errors. The BAO/SN/CMB distance observables are then

$$D_A = \frac{D_M}{1+z}, \qquad D_L = (1+z) D_M, \qquad \frac{dV}{dz\, d\Omega} = \frac{c D_M^2}{H(z)}. \tag{179}$$

The CMB and large-scale-structure layer is transfer-grade rather than completed here:

$$C_\ell^{XY,S^3} = \sum_\beta W_\beta \, \mathcal{P}_\mathcal{R}(\beta) \Delta^X_{\ell\beta} \Delta^Y_{\ell\beta}, \qquad X, Y \in \{T, E\}. \tag{180}$$

For a quotient topology S^3/Γ, the mode catalogue is projected by the deck group:

$$\mathcal{H}_\beta(S^3/\Gamma) = \mathcal{H}_\beta(S^3)^\Gamma, \qquad m^\Gamma_\beta = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi_\beta(g). \tag{181}$$

The full harmonic covariance is then

$$C^{XY,\Gamma}_{\ell m, \ell' m'} = \sum_{\beta, s} \mathcal{P}_\beta \Delta^X_{\ell\beta} \Delta^Y_{\ell'\beta} \xi_{\beta s; \ell m} \xi^*_{\beta s; \ell' m'}. \tag{182}$$

For the round simply connected sector with statistically isotropic perturbations this reduces to the diagonal C_ℓ form. For quotient sectors it can generate off-diagonal covariances and matched-circle-type constraints.

The registered distinguishability targets are likelihood and evidence differences, including the Gaussian covariance KL diagnostic

$$D_{\mathrm{KL}}(i \| 0) = \frac{1}{2} \left[\mathrm{tr}(C_0^{-1} C_i) - N + \ln \frac{\det C_0}{\det C_i} \right]. \tag{183}$$

Until the transfer functions, nuisance parameters, window operators, and covariance are implemented, the compact spectrum remains a mode catalogue and diagnostic registry, not a completed CMB or LSS prediction.

A useful length-to-mode diagnostic is

$$\beta_{\mathrm{wave}}(L_{\mathrm{com}}) = \sqrt{1 + \left(\frac{2\pi R}{L_{\mathrm{com}}} \right)^2}, \qquad \beta_{\mathrm{half}} \simeq \frac{\pi R}{L_{\mathrm{com}}}. \tag{184}$$

For the ring-scale benchmark at $z \simeq 0.8$, the quoted value $\beta \simeq 618.38$ uses the convention that $R_{\mathrm{eff}} = 474.5\,\mathrm{Mpc}$ is a physical shell scale at that redshift and is first converted to a comoving length,

$$\begin{aligned} L_{\mathrm{com}} &= (1+z) R_{\mathrm{eff}} \simeq 2.786\,\mathrm{Gly}, \\ \beta_{\mathrm{half}} &\simeq \frac{\pi R}{L_{\mathrm{com}}} \simeq 618.38. \end{aligned} \tag{185}$$

If 474.5 Mpc is instead treated as an already-comoving full length, the index would be about 1113; if $2R_{\mathrm{eff}}$ is used as the comoving wavelength, it would be about 557. The convention must therefore be fixed before any shell claim is used in a likelihood. This is a registered shell diagnostic, not a Fibonacci law or topology proof.

XXIII Guardrails and Nonclaims

Topology and compactness supply invariant slots, spectra, and covariance structures; they do not supply amplitudes by themselves. In particular, this manuscript does not derive or abolish the dark sector, does not explain late-time acceleration, does not determine dark matter, does not derive Standard Model parameters, and does not claim a preferred cosmic axis. The background expansion $E(z)$ is an external input to the distance and transfer pipeline. The same rule applies to chiral gauge sectors and symmetry breaking. The Hopf fibration makes Hopf and Skyrme topological sectors native on S^3, but it does not quantize the magnetic helicity of an arbitrary primordial $U(1)$ field and does not force nonzero TB or EB CMB spectra. Such parity-odd signatures would belong to a separate chiral gauge-field model with its own action, coupling, initial state, and transfer functions. Likewise, a vector condensate that breaks $SO(4)$ to a subgroup is an additional dynamical hypothesis, not a consequence of the round carrier. Introducing such a field would move the analysis beyond the minimal simply connected S^3_R reconstruction studied here.

The carrier geometry and the stress-energy state are distinct inputs: $T_{\mu\nu}$ is defined on the spacetime geometry and does not by itself select the global topology. All amplitude statements require a dynamics, transfer functions, covariance model, and likelihood comparison.

XXIV Status of Claims

A Mathematical Theorems

The following are exact for any S^3_R:

$$\begin{aligned} D_c &= \pi R, \quad V_{S^3} = 2\pi^2 R^3, \\ K &= \frac{1}{R^2}, \quad {}^{(3)}R = \frac{6}{R^2}. \end{aligned} \tag{186}$$

B Conditional Reconstruction

If a small negative Ω_K is read as a real imprint of positive curvature, then a large finite S^3 radius is recovered. This is conditional on the sign and stability of Ω_K.

C Phenomenological Programme

CMB low-ℓ transfer spectra, BAO/AP distance diagnostics, large-scale structure shell projections, compact harmonic simulations, causal accessibility correlations, and quotient-topology circle or off-diagonal covariance searches.

D What Is *Not* Claimed

- Global S^3 topology is *not* proved.
- The finite curvature radius R or intrinsic diameter πR is *not* a spatial edge or wall.
- The Standard Model of particle physics is *not* derived.
- Perturbation amplitudes are *not* derived.
- The origin of c is *not* proved.
- The φ coordinate is *not* used as a curvature measurement or prediction.
- Temperature-only CMB anomalies are *not* claimed to prove compact topology.
- No tensor-to-scalar amplitude, magnetic-helicity quantization, vector condensate, $SO(4)$ symmetry-breaking field, or Standard-Model anomaly cancellation is derived from topology alone.
- The finite-size Casimir benchmark is *not* claimed to explain dark energy.
- The present reconstruction does *not* derive or replace the late-time acceleration sector; late-time acceleration is treated only through the background function $E(z)$.
- A round simply connected S^3_R is *not* claimed to generate a preferred spatial direction or axis.
- Aharonov–Bohm phases, quantum Hall Chern numbers, non-Abelian anyons, and Chern–Simons invariants are *not* claimed to require a cosmological S^3_R carrier.
- The model is predictive only through the curvature-distance law, transfer functions, and registered likelihood comparisons.

XXV Falsification Conditions

Scientific strength requires explicit falsification criteria. Each component of this programme is falsifiable independently. For clarity, the registered targets are separated from diagnostics:

Target	Observable	Required calculation	Falsification condition
Curvature anchor	$\Omega_K(z)$	$H(z)$ and $D_M(z)$ reconstruction	no constant negative $\Omega_K(z)$ within errors
BAO geometry	D_M/r_d, D_H/r_d, D_V/r_d, $F_{\rm AP}$	same priors/covariance as flat baseline	no likelihood gain or inconsistent sign under joint fit
Compact CMB transfer	TT, TE, EE low-ℓ covariance	hyperspherical Boltzmann transfer sum	no S^3-specific improvement after parameter penalty
Quotient topology	matched circles or off-diagonal $C_{\ell m,\ell' m'}$	group-projected spectrum for S^3/Γ	positive quotient signature rejects $\Gamma = \{1\}$ minimal sector
Shell spectra	$\beta(z, L)$ in LSS catalogues	windowed shell projection with survey mask	no reproducible index across independent masks/catalogues

A Global Curvature Reconstruction

The global closed-S^3 reading rests entirely on a robustly non-zero curvature modulus together with a closed-sign interpretation, $\Omega_K^{S^3} < 0$. Falsification condition: if improved precision from DESI, Euclid, CMB-S4, or successor surveys drives $|\Omega_K| \to 0$, then $R_c \to \infty$ and the finite radius reconstruction loses its observational anchor. If future surveys find $|\Omega_K| < 10^{-4}$ at high significance, or if the reconstructed $\Omega_K(z)$ of Sec. VIII is not redshift-independent within errors, the specific fiducial radius $R_{S^3} = 548\,\mathrm{Gly}$ loses its observational anchor. The compact-mode phenomenology may remain as an independent hypothesis, but the RCP linkage breaks.

B CMB Low-Multipole Prediction

The S^3 Laplacian spectrum supplies a discrete compact-mode catalogue, but for the large fiducial radius used here it does not by itself imply a hard infrared cutoff in the observed low multipoles. The test is therefore not simply whether the quadrupole is low. The test is whether a closed-FLRW hyperspherical transfer calculation predicts a coherent, S^3-specific pattern across low-ℓ TT, TE, and EE spectra, phase correlations, and off-diagonal mode couplings.

Falsification condition: if improved CMB temperature and polarization data, for example from LiteBIRD or CMB-S4, show no detectable S^3-specific low-ℓ pattern under the fiducial curvature radius, the compact-mode CMB explanation is weakened. Temperature-only anomalies are insufficient; polarization is decisive. A concrete likelihood implementation should report

$$\Delta\chi^2_{\rm CMB} = \chi^2_{S^3\text{-transfer}} - \chi^2_{\Lambda\rm CDM} \tag{187}$$

for the registered low-ℓ $TT/TE/EE$ data vector. If the compact transfer model fails to improve the fit, or is disfavoured after the appropriate information-criterion penalty, the CMB compact-mode claim fails as a predictive explanation.

A positive detection of matched circles with a quotient-specific phase relation would favour a multiply connected spherical space form such as S^3/Γ and would falsify the minimal simply connected sector choice, although not the broader positive-curvature compact-geometry programme.

C Ring-Like Structure Predictions

The ring analysis of [2] predicts specific angular scales, redshift-dependent $R_{\rm eff}(z, n, \Delta z, \mathcal{P})$, and a discrete mode selection rule. Falsification conditions:

- DESI/Euclid find no preferred angular clustering at the predicted compact-mode scales in the relevant redshift bins $\Rightarrow$ ring-mode assignment weakened.
- The observed ring scales at $z \approx 0.8$ are better explained by acoustic or projection effects within ΛCDM $\Rightarrow$ S^3-mode interpretation loses parsimony.
- A claimed shell index $\beta(z, L)$ is not reproduced in independent catalogues, redshift shells, or survey masks $\Rightarrow$ projected spectral assignment falsified.

D Compact vs. Box Simulations

Falsification condition: if S^3-harmonic N-body simulations produce no infrared differences from T^3 periodic-box simulations at the observed scales, the compact-geometry simulation programme is weakened.

Summary of falsification logic: Each of the five components is independently testable. Falsification of the global reconstruction ($\Omega_K \to 0$) does not automatically falsify the local spectral programme (ring modes, compact simulations). The programme is *modular*: observational pressure on any one component produces knowledge, not collapse.

Carrier-state distinction. The carrier geometry and the stress-energy state are distinct inputs: $T_{\mu\nu}$ is specified on the spacetime geometry and does not by itself select the global topology.

XXVI Conclusion

If a small curvature modulus is used together with the closed-sign interpretation $\Omega_K^{S^3} < 0$, it specifies a vast but finite S^3 geometry with curvature radius $R \approx 548.3\,\mathrm{Gly}$.

In such a geometry, the observable Universe occupies only a small quasi-local region of approximately 4.82° in particle-horizon radius along the great geodesic of the three-sphere. The locally nearly flat reconstruction is therefore natural, but it does not prove a global $\mathbb{R}^3$ ontology.

The principal scientific claim is:

> The flat infinite framework is not mandatory. The compact S^3 framework preserves local physics while yielding testable global consequences.

This makes the reconstruction scientifically substantive: it does not "invent an age for the Universe", but recovers a possible curvature scale of a compact three-sphere and thereby provides a numerical foundation for further CMB modelling, large-scale structure analysis, and compact spectral cosmology. The CMB low-multipole anomalies do not prove this geometry; they define a disciplined compact-mode falsification target for future temperature and polarization data.

Summary of Key Numerical Values

Table I. Fiducial benchmark numerical values of the S^3_R reconstruction ($R = 548.324\,513\,026\,856\,\mathrm{Gly}$). These machine-precision values are algebraic outputs of the chosen inputs, not observational significant figures. Distances are in gigalight-years (Gly), times in gigayears (Gyr).

Quantity	Symbol	Value	Unit
Curvature radius	$R_{\mathcal{S}^3}$	548.324 513 026 856	Gly
Light-travel time of R	$t_{\mathcal{S}^3}$	548.324 513 026 856	Gyr
Embedding diameter	$D_{\mathrm{embed}} = 2R$	1096.649 026 053 712	Gly
Causal intrinsic diameter	$D_{\mathrm{causal}} = \pi R$	1722.612 261 908 368	Gly
Full great circle	$C = 2\pi R$	3445.224 523 816 736	Gly
Proper volume	$V_{S^3} = 2\pi^2 R^3$	$3.254\,188\,648\,717 \times 10^{9}$	Gly^3
Sectional curvature	$K = 1/R^2$	$3.326\,018\,624\,722 \times 10^{-6}$	Gly^{-2}
Scalar curvature	${}^{(3)}R = 6/R^2$	$1.995\,611\,174\,833 \times 10^{-5}$	Gly^{-2}
Causal accessibility time	$t_{\mathrm{coh}} = \pi R/c$	1722.612 261 908 368	Gyr
Cosmic age	t_0	13.787 718 785 602	Gyr
Age ratio	$t_0/t_{\mathcal{S}^3}$	0.025 145 180 377 75	2.515%
Age as great-circle arc	θ_0	1.440 712 710 740	deg
Particle horizon	D_{particle}	46.125 169 765 204	Gly
Horizon arc	$\theta_{\mathrm{particle}}$	4.819 732 647 519	deg
Volumetric fraction	Γ_{obs}	≈ 0.0126	%
Joint accessibility diagnostic	Γ_{ST}	$4.255\,135 \times 10^{-10}$	–
Boundary-area benchmark	$A_{\partial D_{\mathrm{obs}}}$	$2.667\,234 \times 10^{4}$	Gly^2
Dirac infrared gap	Δ_D	$5.705\,785 \times 10^{-35}$	eV
Conformal scalar Casimir density	$\rho^{(0)}_{\mathrm{Cas}}$	$9.215\,238 \times 10^{-141}$	$\mathrm{J\ m}^{-3}$
Projected shell estimator	$\beta_{\mathrm{shell}}(z = 0.8)$	618.38	convention-dependent
Full-wavelength shell bridge	$\beta_{\mathrm{wave}}(L_{\mathrm{com}})$	$\sqrt{1 + (2\pi R/L_{\mathrm{com}})^2}$	–
Quasi-local scalar kernel	$\xi_{S^3}(\gamma)$	discrete β sum	–

A Derivation of the Scalar Addition Theorem on S^3_R

This appendix derives the scalar addition theorem used in the compact spectral kernel. Let $\mathcal{H}_\beta$ denote the eigenspace of the scalar Laplace–Beltrami operator on the round three-sphere S^3_R with eigenvalue

$$-\Delta_{S^3} Q_{\beta\ell m} = \frac{\beta^2 - 1}{R^2} Q_{\beta\ell m}, \qquad \beta = 1, 2, \ldots, \tag{A1}$$

where $\ell = 0, \ldots, \beta - 1$ and $m = -\ell, \ldots, \ell$. The dimension of this eigenspace is

$$\dim \mathcal{H}_\beta = \beta^2. \tag{A2}$$

For any orthonormal basis $\{Q_{\beta\ell m}\}$ of $\mathcal{H}_\beta$, define the finite-rank spectral projector kernel

$$K_\beta(x, y) = \sum_{\ell=0}^{\beta-1} \sum_{m=-\ell}^{\ell} Q_{\beta\ell m}(x)\, Q^*_{\beta\ell m}(y). \tag{A3}$$

It satisfies the reproducing property

$$\int_{S^3_R} K_\beta(x, z) f(z)\, dV_z = f(x), \qquad f \in \mathcal{H}_\beta. \tag{A4}$$

By $SO(4)$ invariance of the round metric, the kernel can depend on x and y only through their geodesic separation

$$\gamma = \frac{d(x, y)}{R}, \qquad \cos\gamma = \frac{x \cdot y}{R^2}. \tag{A5}$$

Thus $K_\beta(x, y) = K_\beta(\gamma)$ is a zonal harmonic. On S^3, the zonal harmonics of degree n are proportional to Gegenbauer polynomials $C_n^{(1)}(\cos\gamma)$. Since

$$C_n^{(1)}(\cos\gamma) = \frac{\sin[(n+1)\gamma]}{\sin\gamma}, \tag{A6}$$

and the convention used in this manuscript is $\beta = n + 1$, the kernel must have the form

$$K_\beta(\gamma) = A_\beta \frac{\sin(\beta\gamma)}{\sin\gamma}. \tag{A7}$$

The constant A_β is fixed by evaluating the diagonal. Since $K_\beta(x, x)$ is constant by homogeneity,

$$\int_{S^3_R} K_\beta(x, x)\, dV_x = \dim \mathcal{H}_\beta = \beta^2. \tag{A8}$$

Using $\mathrm{Vol}(S^3_R) = 2\pi^2 R^3$, this gives

$$K_\beta(x, x) = \frac{\beta^2}{2\pi^2 R^3}. \tag{A9}$$

Moreover,

$$\lim_{\gamma\to 0} \frac{\sin(\beta\gamma)}{\sin\gamma} = \beta, \tag{A10}$$

so

$$A_\beta = \frac{\beta}{2\pi^2 R^3}. \tag{A11}$$

Therefore the scalar addition theorem on the round three-sphere is

$$\boxed{\sum_{\ell=0}^{\beta-1} \sum_{m=-\ell}^{\ell} Q_{\beta\ell m}(x) Q^*_{\beta\ell m}(y) = \frac{\beta}{2\pi^2 R^3} \frac{\sin(\beta\gamma)}{\sin\gamma}}. \tag{A12}$$

Equivalently,

$$-\Delta_x K_\beta(x,y) = \frac{\beta^2-1}{R^2} K_\beta(x,y), \tag{A13}$$

or

$$\left(-\Delta_x - \frac{\beta^2-1}{R^2}\right) K_\beta(x,y) = 0. \tag{A14}$$

This equation holds for the smooth finite-rank projector K_β. It should not be confused with a Green function: a delta source appears only after summing over all eigenspaces,

$$\sum_{\beta=1}^{\infty} K_\beta(x,y) = \delta_{S^3_R}(x,y) \tag{A15}$$

in the distributional sense.

For the flat-space limit, set $r = R\gamma$ and $k = \beta/R$. At fixed physical separation r and $R \to \infty$,

$$K_\beta(r) \sim \frac{1}{R}\frac{k^2}{2\pi^2}\frac{\sin(kr)}{kr}. \tag{A16}$$

The factor $1/R$ is the discrete spacing Δk. Hence the sum over β becomes the ordinary flat radial Fourier measure,

$$\sum_\beta K_\beta(r) \longrightarrow \int_0^\infty \frac{k^2\,dk}{2\pi^2}\frac{\sin(kr)}{kr}. \tag{A17}$$

This theorem is the exact bridge between the compact spectral catalogue and isotropic two-point statistics on S^3_R. It collapses the sum over (ℓ, m) into a single zonal kernel indexed by β, and it underlies the compact correlation functions, shell projections, windowed responses, mode-coupling matrices, and transfer-function sums used in the main text.

B Complete Invariant Registry

This appendix summarizes the invariant content used in the paper. It is not an encyclopedia of all natural phenomena. It is a closed registry only with respect to the geometric, topological, spectral, observational, and diagnostic structures actually used in the reconstruction.

Class	Invariant object	Formula or criterion	Role in the paper
Local curvature	Sectional curvature	$K = R^{-2}$	Exact geometry
Local curvature	Ricci tensor	$R_{ij} = 2R^{-2} g_{ij}$	Exact geometry
Local curvature	Scalar curvature	${}^{(3)}R = 6R^{-2}$	Exact geometry
Local curvature	Riemann tensor	$R_{ijkl} = R^{-2}(g_{ik}g_{jl} - g_{il}g_{jk})$	Exact geometry
Local curvature	Cotton tensor	$C^{\mathrm{Cotton}}_{ijk} = 0$	Spatial conformal flatness of round carrier
Local symmetry	Covariant curvature derivative	$\nabla_m R_{ijkl} = 0$	Round local symmetry
No-axis theorem	Invariant tangent vector	$(T_p S^3_R)^{SO(3)} = \{0\}$	Excludes geometric preferred axes
Invariant tensors	Symmetric 2-tensor sector	$\mathrm{Sym}^2(T^* S^3_R)^{SO(4)} = \mathbb{R} g$	Excludes geometry-only anisotropic stress
Global geometry	Proper volume	$V_{S^3} = 2\pi^2 R^3$	Compact volume
Global geometry	Intrinsic diameter	$D_c = \pi R$	Maximal geodesic distance
Global geometry	Great circle	$C = 2\pi R$	Closed geodesic scale
Global geometry	Causal-volume identity	$D_c V_{S^3}^{1/3}/R^2 = \pi(2\pi^2)^{1/3}$	Pure round-S^3 benchmark
Geodesic ball	Volume fraction	$\Gamma(\chi) = [\chi - \frac{1}{2}\sin(2\chi)]/\pi$	Observable-domain fraction
Boundary benchmark	Geodesic-sphere area	$A(\chi) = 4\pi R^2 \sin^2\chi$	Isoperimetric diagnostic
Scalar spectrum	Laplace eigenvalues	$\lambda_\beta = (\beta^2 - 1)/R^2$	Compact mode catalogue
Scalar spectrum	Degeneracy	$g^{(0)}_\beta = \beta^2$	Mode counting
Scalar spectrum	Weyl counting law	$N(\Lambda) \sim R^3\Lambda^{3/2}/3$	High-mode flat-limit check
Wave number	Discrete scalar wave number	$k_\beta = \sqrt{\beta^2 - 1}/R$	Bridge to transfer variables
Addition theorem	Scalar two-point kernel	$K_\beta(\gamma) = \dfrac{\beta}{2\pi^2 R^3}\dfrac{\sin(\beta\gamma)}{\sin\gamma}$	Exact correlation kernel
Addition theorem	Kernel normalization	$\int_{S^3} K_\beta\, dV = \delta_{\beta 1}$	Constant-mode/mean-zero separation
Spinors	Untwisted Dirac spectrum	$\lambda^\pm_k = \pm(k + \frac{3}{2})/R$	Infrared spinor gap
Spinors	Dirac degeneracy	$d_k = (k+1)(k+2)$	Spinor mode counting
Topology	Fundamental group	$\pi_1(S^3) = 0$	Simply connected sector
Topology	Cohomology	$H^1 = H^2 = 0,\ H^3 \simeq \mathbb{Z}$	No native one/two cohomology; volume class
Topology	Hopf sector	$S^1 \hookrightarrow S^3 \to S^2,\ \pi_3(S^2) \simeq \mathbb{Z}$	Native Hopf charge domain
Topology	Skyrme degree	$\deg(S^3 \to SU(2) \simeq S^3) \in \mathbb{Z}$	Native Skyrmion topology
Quotients	Deck-group projection	$\mathcal{H}_\beta(S^3/\Gamma) = \mathcal{H}_\beta(S^3)^\Gamma$	Alternative topology sector
Quotients	Mode multiplicity	$m^\Gamma_\beta = \lvert\Gamma\rvert^{-1}\sum_{g\in\Gamma}\chi_\beta(g)$	Quotient spectral selection
FLRW distance	Transverse distance	$D_M(z) = R_c \sin\chi(z)$	Observational curvature law
FLRW test	Curvature consistency	$\Omega_K(z) = ([HD'_M]^2 - c^2)/(H_0^2 D_M^2)$	Reconstruction test
Causal geometry	Accessibility time	$t_{\mathrm{coh}} = \pi R/c$	Global null-accessibility scale
Causal geometry	Accessibility fraction	$\Gamma(T) = [cT/R - \frac{1}{2}\sin(2cT/R)]/\pi$	Causal-domain fraction
Spectral bridge	Full-wave convention	$\beta_{\mathrm{wave}} = \sqrt{1 + (2\pi R/L_{\mathrm{com}})^2}$	Diagnostic length-to-mode map
Spectral bridge	Half-wave convention	$\beta_{\mathrm{half}} \simeq \pi R/L_{\mathrm{com}}$	Registered shell diagnostic
Windowed response	Domain-mode response	$W_\beta[D] = \mathrm{Vol}(D)^{-2}\int_D dV_x \int_D dV_y K_\beta(\gamma)$	Mask/domain diagnostic
Likelihood chain	Predictive endpoint	$\Delta\ln\mathcal{L},\ \Delta\ln Z$	Required status for amplitude predictions
Guardrail	Non-prediction criterion	no likelihood slot $\Rightarrow$ diagnostic only	Prevents overclaiming

The registry separates exact mathematical statements from observational claims. Objects in the first columns are exact within the chosen round-S^3_R model. Their physical use becomes predictive only after they enter an observable transfer or likelihood pipeline. The causal-volume identity and the shell-index rows are benchmarks: they are not independent detections of topology. This is the operative closure rule of the paper:

$$\begin{aligned} S^3_R &\longrightarrow \{D_M, H, dV/dz\, d\Omega, k_\beta, K_\beta\} \\ &\longrightarrow \{C_\ell^{TT,TE,EE}, P(k,z), \xi(r,z)\} \\ &\longrightarrow \Delta\ln\mathcal{L} \text{ or } \Delta\ln Z. \end{aligned} \tag{B1}$$

Anything outside this chain is retained only as a diagnostic, a taxonomy entry, or an explicit non-claim.

C Optional Descriptive φ Scale Coordinate

The φ-hierarchy is not used as evidence for the compact reconstruction. It is a descriptive scale coordinate only:

$$n_\varphi(R) = \frac{\ln(R/\ell_P)}{\ln \varphi}, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{C1}$$

For the fiducial radius this gives $n_\varphi(R_{\mathcal{S}^3}) \simeq 299.09$. This number is not a topology proof, not a curvature measurement, and not an amplitude prediction. The established appearance of φ in Fibonacci anyon theory is a mathematical analogy in topological quantum matter, not an independent derivation of cosmological curvature.

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